



Foundations of Programming

Lab Assignment No. : 03

PROBLEM STATEMENT:

Write a program in C to generate multiplication tables for a given number.

OBJECTIVES:

- To understand the use of looping statements in C.
- To generate multiplication tables using loops.
- To improve program logic and execution flow understanding.

THEORY:

Syntax for for loop example program

```
for (initialization; condition; increment/decrement) {  
    // statements  
}
```

```
#include <stdio.h>
```

```
int main() {
```

```
    int i;
```

```
    for (i = 1; i <= 5; i++) {
```

```
        printf("%d ", i);
```

```
}
```

```
return 0;
```

```
}
```

Syntax for while loop example program

```
initialization;
```

```
while (condition) {
```

```
    // statements
```

```
    increment/decrement;
```

```
}
```

```
#include <stdio.h>
```

```
int main() {
```

```
    int i = 1;
```

```
    while (i <= 5) {
```

```
        printf("%d ", i);
```

```
        i++;
```

```
    }
```

```
    return 0;
```

```
}
```

Syntax for do while loop example program

```
initialization;
```

```
do {  
    // statements  
    increment/decrement;  
} while (condition);  
#include <stdio.h>
```

```
int main() {  
    int i = 1;  
  
    do {  
        printf("%d ", i);  
        i++;  
    } while (i <= 5);  
  
    return 0;  
}
```

ALGORITHM:

Step 1: Start

Step 2: Declare variables num and i

Step 3: Read a number from the user

Step 4: Set i = 1

Step 5: Repeat while i <= 10

 Display num × i = result

 Increase i by 1

Step 6: Stop

Type Program (Multiplication table)

```
#include <stdio.h>
```

```
int main() {  
    int num, i;  
  
    printf("Enter a number: ");  
    scanf("%d", &num);  
  
    printf("Multiplication Table of %d:\n", num);  
  
    for (i = 1; i <= 10; i++) {  
        printf("%d x %d = %d\n", num, i, num * i);  
    }  
  
    return 0;  
}
```

```
1  #include <stdio.h>
2
3  int main() {
4      int num, i;
5
6      printf("Enter a number: ");
7      scanf("%d", &num);
8
9      printf("Multiplication Table of %d:\n", num);
10
11     for (i = 1; i <= 10; i++) {
12         printf("%d x %d = %d\n", num, i, num * i);
13     }
14
15     return 0;
16 }
17
```

PLATFORM:

Linux

INPUT:

Choice of operation and two numbers

OUTPUT:

Displays the result of the selected arithmetic operation

SAMPLE INPUT:

5

SAMPLE OUTPUT:

```
Enter a number: 5
Multiplication Table of 5:
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
5 x 6 = 30
5 x 7 = 35
5 x 8 = 40
5 x 9 = 45
5 x 10 = 50
```

CONCLUSION:

Thus, the menu driven program in C successfully performs basic arithmetic operations using switch-case statements.

FAQs:

1. Which loop is best suited for generating multiplication tables?
2. What is the role of loop initialization?
3. Difference between for loop, while loop and do while loop?
4. Can a do-while loop be used to generate multiplication tables? Explain.
5. Why is the do-while loop called an exit-controlled loop?